Variance Explanation Assistant

Product Brief | Track 4: Finance | Author: Anuj Patel

Public app: variance-assistant--anujpatel999.replit.app
Code: github.com/anujpat999-art/variance-assistant

1. Problem Statement

At month-end, leadership looks at a forecast-versus-actual table and asks one question: *why did we miss?* The finance analyst has the numbers but has to produce a defensible explanation quickly. In practice this means hunting through category lines, working out which ones actually matter, separating price from volume effects, and writing a summary, often under time pressure and with the answer needed in the same meeting.

Two things go wrong. First, **effort goes to the wrong lines**: analysts explain every small variance equally, while two or three lines usually account for most of the miss. Second, **explanations are written from memory or assumption**, so a cause gets stated as fact when the data does not support it. When a leader challenges it, the analyst cannot defend it. The cost is lost credibility and a rework cycle after the review.

This product helps the analyst find the lines that matter, calculate the figures correctly, and draft an explanation that is tied to evidence, with anything unsupported marked as unclear.

2. User Persona

Representative persona based on the track brief; not from user interviews.

Who	Finance analyst (FP&A or business finance) at a mid-sized company, 2-5 years of experience.
Context	Monthly close. Owns the variance commentary that goes to the CFO or business head. Works in spreadsheets, with data exported from an ERP or BI tool.
Trigger moment	Month-end review: leadership asks "Why did we miss the forecast?" and the analyst has little time to answer.
Goals	Explain the miss clearly, show it is backed by numbers, and be ready for follow-up questions.
Frustrations	Too many lines to explain; price vs volume split is manual; fear of stating a cause that turns out to be wrong; commentary rewritten each month.
What trust means	Every figure traceable to the input. Honest "we cannot tell from this data" is better than a confident guess.

3. JTBD Statement

When I see variance in numbers, **I want to** explain what caused it clearly and quickly, **so I can** defend my analysis.

Success looks like: the analyst can walk into the review with a short, correct explanation of the few drivers that matter, each tagged as volume, price, timing or anomaly, with a clear statement of what is still unknown.

4. Why AI? (Critical Thinking)

Much of this job does **not** need AI, so the product splits the work deliberately.

Task	Done by	Reason
Variance, % variance, share of total, favorable or unfavorable	Code	Arithmetic must be exact and repeatable. An LLM can get sums wrong.
Which lines are material	Code (thresholds)	A transparent rule the analyst can adjust and defend.

Task	Done by	Reason
Price vs volume split	Code	Exact identity: actual units x actual price - forecast units x forecast price = volume effect + price effect.
Obvious causes from structured evidence (notes mention delay or one-off, statistical outlier)	Code (rules)	Cheap, explainable, no model needed.
Turning figures, rule hints and analyst notes into a concise executive explanation	**LLM**	Language generation and prioritising what to say is where a model adds value.
Spotting what a CFO will ask next	**LLM**	Open-ended reasoning over the evidence; produces follow-up questions.

What AI is not trusted with: calculating numbers, deciding causes without evidence, or classifying a driver as volume or price when no unit or price data exists. The logic layer runs first, and the model receives only computed facts. A rules-only version would produce correct numbers but stiff, repetitive commentary; an AI-only version would be fluent but unreliable. The combination is the product.

5. Feature List + Value vs Effort

Feature	User value	Effort	Decision
Structured input form (category, type, forecast, actual, optional units/price, note)	High	Low	Built
Input validation with clear error messages	High	Low	Built
Variance %, share of total, favorable/unfavorable, materiality filter	High	Low	Built
Price-volume split when units and prices are given	High	Medium	Built
Rule-based classification hints (timing, one-off, outlier)	Medium	Low	Built
LLM explanation per driver + executive summary + follow-up questions	High	Medium	Built
Grounding checks and confidence score	High	Medium	Built
Saved reports and history	Medium	Low	Built
Admin dashboard (success rate, confidence, cost, classes, flags, feedback)	Medium (for the builder/operator)	Medium	Built
CSV paste import	Medium	Low	Built
Direct ERP / spreadsheet connectors	High	High	Later
Multi-period trend and mix variance	Medium	High	Later
User accounts and team sharing	Medium	Medium	Later
Slide or PDF export	Low	Low	Skipped (browser print works)

Priority was high value for low effort first, and everything that protects trust (validation, grounding, confidence) before anything that adds convenience.

6. Failure Mode Analysis

The key failure mode named in the brief is **inventing causes not supported by the data**. It is the one that destroys trust, because a finance leader who catches one made-up cause stops believing the rest.

Failure mode	How it shows up	Mitigation in the product
Invented cause	Model says "supplier price increase" with no such evidence	Prompt forbids it; volume/price rejected unless unit and price data exist; timing rejected unless the note says so; downgraded to **unclear** with a visible note
Invented or wrong numbers	Model quotes a figure not in the input	Numbers in AI text are compared with supplied figures; mismatch is flagged and confidence is capped
Over-explaining small lines	Noise buries the real drivers	Materiality rule (min % and min share of total); only top drivers sent to the AI; nothing material means no AI call
Overconfidence	Fluent text reads as certain	Confidence blends the model's rating (60%) with an evidence score from the logic layer (40%); capped when top drivers explain under 60% of the variance; unclear drivers capped
Bad or messy input	Blank period, text in a number field, duplicates, negative values	Validation returns specific messages and keeps the form filled
LLM outage, missing key or malformed output	API error, timeout, no API key, invalid JSON	One retry; on failure the report still saves with calculated figures and a visible notice
Model skips or adds drivers	Output does not match the input categories	Unknown categories dropped; skipped drivers flagged
Wrong sign convention	Revenue up is good, expense up is bad	Each line is typed revenue or expense; favorable/unfavorable computed in code

Residual risks: the tool cannot know business context the analyst did not enter, and the grounding check only catches numbers of 100 or more or percentages, so small invented figures could slip through. The analyst remains responsible for the final commentary; the tool is a drafting aid.

7. AI Trade-offs

Decision	Chosen	Alternative and why not
Where logic lives	Deterministic code before the LLM	LLM does everything: simpler, but unreliable math and untraceable reasoning.
What the model sees	Computed facts for top drivers only	Send the raw table: more tokens, more room to invent, more cost.
Output format	Strict JSON, parsed and validated, shown as a structured report	Free text: easier to prompt, impossible to check or store reliably.
Model size	Small, low-cost model (default gpt-4o-mini, configurable; Anthropic supported)	Largest model: better prose, but cost and latency rise for a task that is mostly constrained summarisation.
Temperature	Low (0.2)	Higher creativity is a liability here.
Strictness	Prefer "unclear" over a guess	Fewer confident answers, but each one is defensible. Some users may find it conservative.
Privacy	Only category-level figures and short notes are sent	Sending full ledgers would expose more sensitive data.

Latency is a few seconds per analysis, which is acceptable for a month-end task. The model-provider choice is a single environment setting, so it can be switched without code changes.

8. Cost & Sustainability

Per-analysis cost. A typical request is roughly 1,000-2,000 input tokens and 400-800 output tokens. At small-model pricing (about $0.15 per million input and $0.60 per million output tokens for gpt-4o-mini, to be

checked against current pricing), that is well under one US cent per analysis, around a few hundredths of a cent. The admin dashboard records actual tokens and estimated cost per report, so this estimate can be replaced with measured data.

Cost controls built in:

- The AI is skipped entirely when nothing is material.
- Only the top 5 drivers are sent, never the full table.
- A small model, low token limit, and one retry only.
- Calculations are free code, not model calls.

Sustainability of the product. Hosting is a small web app with SQLite, cheap or free to run for a prototype. To scale, the first moves are switching to a hosted database, adding authentication, and rate limits. If the product charged per seat, the model cost per analyst per month would be a small fraction of any realistic subscription. The main ongoing cost is maintenance: keeping prompts, pricing tables and guardrails current as models change.

Environmental note: using a small model and avoiding unnecessary calls also keeps energy use per analysis low.

9. Reflection

The main lesson was that the AI is the smaller half of this product. The value came from deciding what code should do before the model is involved: exact variance maths, materiality, and a price-volume split. That also made the AI safer, because it receives facts instead of a raw table and has fewer chances to invent something.

The hardest design question was how to handle uncertainty. A tool that always gives a confident cause would look impressive in a demo and fail the first time a CFO pushes back. Allowing the answer "unclear, and here is what would confirm it" makes the tool less flashy but more trustworthy.

If I continued, I would test it with real analysts on real month-end data, measure how often they edit the output, and use the helpful / not helpful feedback and the dashboard's flag counts to tighten the guardrails. I would also add spreadsheet import, since typing categories by hand is the biggest friction in the current prototype.

Limitations: persona is representative rather than interviewed; confidence is a heuristic blend, not a calibrated probability; and only single-period, category-level variance is supported. The public deployment was run without a live LLM API key, so it shows the calculated analysis with the AI explanation step falling back gracefully; the AI integration is built and covered by tests using a mocked model, but has not been exercised against a live model.